

Substring Reversals

Time limit: 1.00 s

Memory limit: 512 MB

Given a string, your task is to process operations where you reverse a substring of the string. What is the final string after all the operations?

Input

The first input line has two integers n and m : the length of the string and the number of operations. The characters of the string are numbered $1, 2, \dots, n$.

The next line has a string of length n that consists of characters A–Z.

Finally, there are m lines that describe the operations. Each line has two integers a and b : you reverse a substring from position a to position b .

Output

Print the final string after all the operations.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $1 \leq a \leq b \leq n$

Example

Input	Output
7 2 AYBABTU 3 4 4 7	AYAUTBB